FEDERAL SPECIFICATION

INSECTICIDE, RESMETHRIN (AEROSOL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of aerosol insecticide formulation containing resmethrin (synthetic pyrethroid) in a 12-ounce dispenser complete with valve.

1.2 Classification. The aerosol insecticide formulation shall be of the following types as specified (see 6.2):

Type I - 1.2 percent resmethrin.
Type II - 2.0 percent resmethrin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

BB-F-1421 - Fluorocarbon Refrigerants.
NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
~~PPP-F-320~~ - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" by 48" Pallets.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 171 to 179 - ~~Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.~~

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(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

Federal Insecticide, Fungicide, and Rodenticide Act.

Title 7 United States Code, Section 135 et seq. as amended by the Federal Environmental Pesticide Control Act of 1972 (PL 92-516) and any regulations promulgated by the Administrator of the Environmental Protection Agency.

(Application for copies should be addressed to the Environmental Protection Agency, Pesticide Regulation Division, 401 M St., S.W., Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

D1193 - Reagent Water.
D3069 - Delivery Rate of Aerosol Products.
D3088 - Effectiveness of Aerosol and Pressurized Space Spray Insecticides Against Flying Insects.
D3092 - Particle-Size Distribution of Space-Insecticide Aerosol.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

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Soap and Sanitary Chemicals, Vol. XXIX, No. 2, February, 1953.

(Soap and Sanitary Chemicals is published by the MacNair-Dorland Company, 101 West 31st Street, New York, NY 10001.)

3. REQUIREMENTS

3.1 Aerosol insecticide formulation composition. The aerosol insecticide formulation shall conform to the applicable composition of table I (see 4.2.3.1). No other ingredients shall be used in the aerosol insecticide formulation.

TABLE I. Composition of aerosol insecticide formulation

Ingredient	Percent by weight	
	Type I	Type II
Resmethrin, actual (5-benzyl-3-furyl)methyl 2,2-dimethyl-3-(2-methylpropenyl)cyclopropanecarboxylate from premium technical grade conforming to 3.1.1, minimum	1.2	2.0
Dichlorodifluoromethane conforming to type 12 of BB-F-1421	49.0 ± 1.0	49.0 ± 1.0
Trichloromonofluoromethane conforming to type 11 of BB-F-1421	49.0 ± 1.0	49.0 ± 1.0

3.1.1 Resmethrin, premium technical grade. The resmethrin used in the aerosol insecticide formulation shall conform to the chemical characteristics of table II when tested as specified therein. The supplier shall certify that the stabilizer requirement has been met.

TABLE II. Chemical characteristics of premium technical grade resmethrin

Characteristic	Percent by weight	Test paragraph
(5-Benzyl-3-furyl)methyl 2,2-dimethyl-3-(2-methylpropenyl)cyclopropanecarboxylate, minimum	96.0	4.2.4.1
Cis/trans isomer ratio of above ingredient:		4.2.4.2
(±)cis isomer, maximum 11.0 percent by weight		
(±)trans isomer, minimum 89.0 percent by weight		
2,6-Di-tertiary-butyl-para-cresol (stabilizer)	3.5 ± 0.5	-------

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3.2 Resmethrin content in aerosol insecticide formulation. Type I aerosol insecticide formulation shall contain no less than 1.2 percent by weight resmethrin when tested as specified in 4.2.4.3. Type II aerosol insecticide formulation shall contain no less than 2.0 percent by weight resmethrin when tested as specified in 4.2.4.3.

3.3 Water content. After the dispenser has been filled with aerosol insecticide formulation, the formulation shall contain no more than 80 parts per million of water when tested as specified in 4.2.4.4 (see 6.3).

3.4 Net weight of contents. Each dispenser shall contain 12.0 (+ 0.2 or -0) ounces of aerosol insecticide formulation when tested as specified in 4.2.4.3(c).

3.5 Effectiveness against flying insects. The aerosol insecticide formulation shall pass the test for effectiveness against flying insects when tested as specified in 4.2.4.5.

3.6 Dispenser and valve. The dispenser container shall be a metal can conforming to type IX, class 2 or 3 of PPP-C-96. The container shall have a capacity of no less than 25 cubic inches. The container shall be supplied with a dispensing valve having the valve performance characteristics specified in table III. The valve system shall conform to one of the following three models:

(a) Precision Valve Company, 700 Nepperhan Avenue, Yonkers, NY 10703
Valve: N,N No. 1
Actuator: 01-1406, 0.018 inch, polypropylene with arrow
Stem: 04-1210-04, 0.013 inch radius cored
Stem gasket: 05-0310, "Buna N"
Spring: 06-6010, stainless steel
Body: 07-1901, 0.080 inch pressure filling
Mounting cup: 12-1230, conical, splined, undimpled with flowed-in gasket
Dip tube: 09-2010, notched PVC, proper length to empty dispenser container contents

(b) Seaquist Valve Co., 1160 North Silver Lake Road, Cary, IL 60013
Valve: Seaspray NS-31
Stem: 0.016 inch
Mounting cup regular: 1 inch
Gasket: neoprene
Spring: 0.023 inch wire, stainless steel
Body orifice: 0.018 inch
Vapor tap: none
Capillary inside diameter: none
Actuator: Arrow #0100-0102-15, 0.015 inch reverse taper
Dip tube: proper length to empty dispenser container contents

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(c) Valve Corporation of America, 1720 Fairfield Avenue, Bridgeport, CT 06605
Valve: B-9-FG
Stem: 0.014 inch
Gasket: neoprene
Pressure filled
Housing: 0.100 inch
Plain finish
Dip tube size: 0.160 inch inside diameter, proper length to empty dispenser container contents
Actuator type: B-9-10T

The valve shall be fitted with a finger operated side discharge plastic button. The direction of discharge shall be indicated either by the shape of the button or by a mark on it. The valve and button shall be constructed of materials which are not affected by the formulation.

TABLE III. Valve performance characteristics

Characteristic	Requirement	Test paragraph
Delivery rate at 80° F	1.05 ± 0.30 g per second	4.2.4.6
Particle size distribution:		4.2.4.7
Up to 30 microns, minimum	80.0 percent	
30 to 50 microns, maximum	20.0 percent	
Over 50 microns	0.0 percent	
Leakage in 48 hours, maximum	0.5 ml	4.2.4.8

3.7 Protective cap. The dispensing valve shall be protected against actuation by a crown cap as specified in PPP-C-96. The cap shall provide a clearance of no less than 1/8 inch directly over the valve button. The cap shall provide a snug fit and shall completely cover the end of the dispenser containing the valve assembly.

3.8 Workmanship. The aerosol insecticide formulation shall be mixed by a suitable process to produce a homogeneous mixture. The dispenser shall be clean and uniform and free of any defects which might impari its utility such as sediment, metal particles, or other foreign material in the container, education tube, or valve assembly. The exterior orifice shall be symmetrical and free of ragged edges. The exterior orifice, if drilled, shall be symmetrical in direct alignment with the angle of discharge.

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3.9 Labeling. Labeling data shall be legibly lithographed on the dispenser container with weather-resistant noncorrosive ink (black on clear-coated containers and contrasting color on color-coated containers). Paper labels shall not be used. Each dispenser shall be labeled in compliance with the Federal Insecticide, Fungicide, and Rodenticide Act. The label shall be accepted by and registered with the Pesticide Regulation Division of the Environmental Protection Agency. The label shall include the name and percent by weight of each active ingredient, the name and percent by weight of each inactive ingredient, net weight of contents, stock number, type designation, cis/trans ratio, and U. S. patent number.

3.9.1 Required label information.

3.9.1.1 Type I. The type I aerosol insecticide formulation label shall contain, as a minimum, the following directions for use and precautionary information:

WARNING: KEEP OUT OF REACH OF CHILDREN

Warning - Contents under pressure. Do not puncture. Do not store near heat or open flame. Exposure to temperatures above 130° F may cause bursting. Never throw container into fire or incinerator.

NOTICE
THE INGREDIENTS IN THIS PRODUCT ARE NON-FLAMMABLE
AND THE SPRAY MIST WILL NOT IGNITE

FOR USE AGAINST MOSQUITOES, FLIES, AND GNATS (MIDGES)

Directions for use

Statement of Use Classification (General or Restricted)

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

Remove protective cap. Aim spray opening away from person. Push button to spray.

INDOOR AREAS - Close space to be treated. Release 3 feet from any surface, holding the dispenser above the head. Moving the dispenser about while spraying aids in dispersion. Dispense only for time indicated. Five seconds of spraying is ample for each 1,000 cubic feet (10 feet by 10 feet by 10 feet). It is not necessary to spray directly on insects but merely spray into the air since the aerosol will disperse throughout the enclosed space. Keep treated area closed for at least 30 minutes. Ventilate area before re-entry. Repeat as necessary.

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SHELTERED AREAS - In such areas that can't be closed tightly, close in best possible way. Fill entire area with mist. Keep treated area closed for at least 30 minutes. Ventilate area before re-entry. Repeat as necessary.

CAUTION - Harmful if swallowed. Avoid inhalation of aerosol mist. Avoid contamination of feed and foodstuffs. Do not use on edible crops. Foods should be removed or covered during treatment. All food processing surfaces should be covered during treatment or thoroughly cleaned before using. When using the product in these areas apply only when the facility is not in operation. Remove pets and birds. Cover or remove aquariums. Never spray directly on plastics.

3.9.1.2 Type II. The type II aerosol insecticide formulation label shall contain, as a minimum, the following directions for use and precautionary information:

NOTICE
THE INGREDIENTS IN THIS PRODUCT ARE NON-FLAMMABLE
AND THE SPRAY MIST WILL NOT IGNITE

FOR PUBLIC HEALTH, PLANT PROTECTION AND QUARANTINE PESTS FOUND ON AIRCRAFT. KILLS FLIES, MOSQUITOES, GNATS, AND CARIBBEAN FRUIT FLIES.

Directions for use

Statement of Use Classification (General or Restricted)

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

Disinsectization may be accomplished either while the aircraft is in flight or while on the ground prior to takeoff, but in no case shall disinsectization be accomplished later than 30 minutes prior to the first landing at a United States port. The ventilation system shall be stopped and all the openings to the exterior kept closed while the insecticide is being released or sprayed, and for a period of not less than 3 minutes thereafter. If disinsectization is accomplished on the ground prior to takeoff, the ventilation system may be opened at the end of such period, provided that the system is equipped with filters or other means for preventing the entrance of insects into the aircraft but doors, windows, or other such openings to the exterior shall be kept closed until after the takeoff, except in the case of emergency, or to permit persons applying the insecticide to disembark. The aerosol should be dispersed at blocks away (as the aircraft is taxiing from the terminal) at a dosage of 5 grams (5 seconds) per 1,000 cubic feet of space. Remove protective cap. Direct the spray nozzle toward

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the ceiling of the compartment or space being sprayed. Push button to spray. Move arm from side to side in a sweeping motion. Spray the spaces inaccessible from within aircraft after completely loading fuel, passengers, cargo, baggage, etc; this includes baggage compartments, wheel wells and other similar places. Spray the cabin, cockpit and other spaces accessible from within the aircraft after the crew is on board and after all doors, windows, hatches and ventilation openings are closed.

WARNING: KEEP OUT OF REACH OF CHILDREN

WARNING: Contents under pressure. Do not puncture. Do not store near heat or open flame. Exposure to temperatures above 130° F may cause bursting. Never throw container into fire or incinerator.

CAUTION

Harmful if swallowed. Avoid contamination of feed and foodstuffs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible fo for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the perfcrmance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dispensers that have been filled with aerosol insecticide formulation of one type which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.2.2 Sampling.

4.2.2.1 For examination of filled dispensers. Sampling for examination of filled dispensers shall be conducted in accordance with MIL-STD-105, inspection level I, using an AQL of 2.5 percent defective.

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4.2.2.2 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105, inspection level S-2 using an AQL of 4.0 percent defective.

4.2.2.3 For test. Sampling for test for net weight of contents, delivery rate at 80° F, particle size distribution, and leakage shall be conducted in accordance with MIL-STD-105, inspection level S-3 using an AQL of 2.5 percent defective. Six additional samples shall be selected for resmethrin content and water content tests; three for each test.

4.2.3 Inspection procedure.

4.2.3.1 For composition of aerosol insecticide formulation. Weight records of the amount and lot number of each ingredient used to produce each lot of aerosol insecticide formulation shall be kept for the information of an inspector.

4.2.3.2 For examination of filled dispensers. The sample unit shall be one filled dispenser. Sample dispensers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Dispenser corroded, bent or distorted
(b) Dispenser container not as specified
(c) Valve discharge button not as specified
(d) Protective cap not as specified or not a snug fit
(e) Labeling incorrect, missing, or illegible
(f) Workmanship not as specified (see 3.8)

4.2.3.3 For examination of preparation for delivery. The sample unit shall be one filled shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container or arrangement thereof not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard packing aids not as specified
(f) Strapping not as specified
(g) Marking incorrect, missing, or illegible

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4.2.3.4 For test. Sample dispensers taken in accordance with MIL-STD-105 in 4.2.2.3 shall be tested for net weight of contents, delivery rate at 80° F, particle size distribution, and leakage using an AQL of 2.5 percent defective. Three of the six additional samples taken shall be tested for resmethrin content; the remaining three samples shall be tested for water content. Failure of any of the sample dispensers to pass the resmethrin content test or the water content test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Resmethrin assay of premium technical grade resmethrin.

(a) Apparatus. Use a gas chromatograph equipped with a flame ionization detector and a suitable integrating recorder. Recommended chromatographic conditions are shown in table IV. These conditions may be modified, as necessary, for the gas chromatograph used to yield symmetrical peaks with minimum tailing and base line separation.

TABLE IV. Chromatographic conditions

Characteristic	Condition
Column material	Glass
Column dimensions	6 ft by 2 mm ID
Column support	80 to 100 mesh "Chromasorb W(HP)"
Column coating	10% "OV-1"
Injector temperature	250° C
Column temperature	250° C
Detector temperature	250° C
Carrier gas	Helium
Carrier gas flow rate, ml per min	30

(b) Internal standard solution. Dissolve 3.5 grams (g) of docosane in sufficient chloroform to make 250.0 milliliters (ml).

(c) Preparation of standard solution. Transfer approximately 200 milligrams (mg) of analytical standard resmethrin (see 6.5) which has been accurately weighed into a 25-ml volumetric flask and dissolve in about 10 ml of chloroform. Add 5.0 ml of the internal standard solution prepared in (b), dilute to the volume mark with chloroform, and mix.

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(d) Preparation of specimen solution. Dissolve the specimen by gently warming on a steam bath and mixing thoroughly. Using a syringe, transfer about 0.2 ml of the specimen into a tared 25-ml volumetric flask, immediately stopper, and accurately weigh the flask. Determine the weight of the specimen by difference. Add 5.0 ml of the internal standard solution prepared in (b), dilute to the volume mark with chloroform, and mix.

(e) Procedure. Condition the column before use by repetitively injecting the standard solution prepared in (c). The column is stabilized when the peak height ratios of the first peak (docosane) and the second peak (resmethrin) are constant. Inject 1.0-microliter portions of the standard solution prepared in (c) and the specimen solution prepared in (d) alternately and repetitively until sufficient chromatograms of each are obtained. Calculate the average response factor of resmethrin relative to docosane from the chromatograms of the standard solution as follows:

$$\text{Average response factor} = \frac{AB}{CDE}$$

where: A = Weight of docosane in standard solution in milligrams,
B = Area of standard resmethrin peak,
C = Percent purity of standard resmethrin,
D = Weight of standard resmethrin in standard solution in milligrams, and
E = Area of docosane peak.

Calculate the percent by weight resmethrin from the chromatograms of the specimen solution as follows:

$$\text{Percent resmethrin} = \frac{100\ FG}{HJK}$$

where: F = Area of resmethrin peak,
G = Weight of docosane in specimen solution in milligrams,
H = Area of docosane peak,
J = Weight of specimen in specimen solution, and
K = Average response factor calculated above.

4.2.4.2 Cis/trans isomer ratio of technical grade resmethrin.

(a) Apparatus. Use a gas chromatograph equipped with a flame ionization detector and a suitable recorder. Recommended chromatographic conditions are shown in table V. These conditions may be modified, as necessary, for the gas chromatograph used to yield symmetrical peaks with minimum tailing and base line separation.

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TABLE V. Chromatographic conditions

Characteristic	Condition
Column material	Glass
Column dimensions	12 ft by 4 mm ID
Column support	100 to 120 mesh "Gas-chrom Q"
Column coating	3% "OV-1"
Injector temperature	210° C
Column temperature	210° C
Detector temperature	210° C
Carrier gas	Helium
Carrier gas flow rate, ml per min	70

(b) Procedure. Dissolve the specimen by gently warming on a steam bath and mix thoroughly. Using a syringe, transfer 0.5 ml to a 25-ml volumetric flask. Dilute to the volume mark with chloroform and mix. Condition the column before use by injecting several portions of the specimen in chloroform solution. When the column is stabilized inject 1.0 microliter of the solution into the chromatograph and measure the area under the first peak (cis isomer) and the area under the second peak (trans isomer). Calculate the percent by weight cis isomer and trans isomer as follows:

$$\text{Percent cis isomer} = \frac{100A}{C}$$

$$\text{Percent trans isomer} = \frac{100B}{C}$$

where: A = Area under the cis peak,
B = Area under the trans peak, and
C = Area under the cis peak plus area under the trans peak.

4.2.4.3 Resmethrin content in aerosol insecticide formulation.

(a) Apparatus. Use a gas chromatograph equipped with a thermal conductivity detector. Recommended chromatographic conditions are shown in table VI. These conditions may be modified, as necessary, to obtain optimum response and reproducibility.

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TABLE VI. Chromatographic conditions

Characteristic	Condition
Column dimensions	4 feet by 1/4 inch ID
Column support	80 to 100 mesh "Chromasorb 750"
Column coating	10% "SP-2100"
Column temperature	260° C
Injection temperature	290° C
Detector temperature	270° C
Filament current, milliamps	200
Carrier gas	Helium
Carrier gas flow rate, ml per min	55

(b) Preparation of standard solution. Weigh 0.06 g of analytical standard resmethrin (see 6.5) into a 10-ml volumetric flask, dissolve in pesticide or spectro grade carbon disulfide and dilute to the volume mark with carbon disulfide. Add a small amount of anhydrous sodium sulfate to insure dryness.

(c) Preparation of specimen solution and net weight of contents determination. Record the gross weight of the filled sample dispenser container and cap. Place the dispenser container in a freezer overnight or in a dry ice chest for at least 2 hours. Remove the dispenser container and immediately punch several holes in the top to relieve pressure. Open the dispenser container top with a can opener and allow to warm to room temperature. Remove any remaining propellants by placing the dispenser container in a water bath at 35° to 40° C. Dissolve the residue, transfer to a 100-ml volumetric flask, and dilute to the volume mark with pesticide or spectro grade carbon disulfide. Dry the dispenser container and weigh it with its cap. Subtract this weight from the gross weight to obtain the net weight, which is both the net weight of contents and the weight of the specimen in this test. Using the declared percent by weight resmethrin and the weight of the specimen, calculate the apparent concentration of resmethrin in the 100-ml volumetric flask. Dilute an aliquot of the solution to obtain a solution having a resmethrin concentration of approximately 6 mg per milliliter.

(d) Procedure. Using a 25-microliter precision liquid syringe, alternately inject three 10 to 20 microliter portions of the standard solution prepared in (b) and the specimen solution prepared in (c). Measure the peak area for each peak and calculate the average peak area for both the standard solution and the specimen solution.

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Adjustments in attenuation or amount of solution injected may be necessary to give convenient size peaks. Calculate the percent by weight resmethrin in the aerosol insecticide formulation as follows:

$$\text{Percent resmethrin} = \frac{ABC}{DE}$$

where: A = Average peak area of specimen solution,
B = Weight of standard resmethrin injected in milligrams,
C = Percent purity of the standard resmethrin,
D = Average peak area of specimen solution, and
E = Weight of specimen injected in milligrams.

4.2.4.4 Water content. Determine water content in accordance with the method described by R. A. Fulton and R. P. Gelardo in "A Method for Determining Water in Insecticidal Aerosols," Soap and Sanitary Chemicals, Volume XXIX, No. 2, pp. 145, 147, and 159, February, 1953.

4.2.4.5 Effectiveness against flying insects. Determine effectiveness against flying insects in accordance with ASTM D3088.

4.2.4.6 Delivery rate at 80° F. Determine delivery rate in accordance with ASTM D3069 using a water bath maintained at 80° ± 0.5° F.

4.2.4.7 Particle size distribution. Determine the particle size distribution in accordance with ASTM D3092.

4.2.4.8 Leakage. Dispense ten separate 1-second applications of the aerosol insecticide formulation from each dispenser container selected for leakage testing. Then immerse the dispenser container in a deep beaker of water and invert a 1-ounce, wide-mouth vial, calibrated to read in milliliters, full of water over the valve of the container and allow to stand for 48 hours at 25° ± 2° C. Record the gas volume in the calibrated vial as the amount of leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Twenty-four filled dispensers shall be packed upright in six rows of four dispensers per row in a snug-fitting box conforming to class 2, style 4-1/2, grade A of PPP-B-621; type overseas, style I or J, grade A of PPP-B-601; or class 2, style 2 of PPP-B-585. The box shall be designed for a type 2 load. Each dispenser shall be placed in a snug-fitting cell formed by use of half-slotted fiberboard partitions of the same height as the dispenser ± 1/8 inch. All inside faces of the box shall be lined with fiberboard. Additional fiberboard sheets

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shall be added where needed to assure a tight pack. The fiberboard partitions and liners shall be formed from material conforming to type CF, class weather-resistant, variety SW, grade W5c or W6c of PPP-F-320. The filled box shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification.

5.1.2 Level B.

5.1.2.1 For military activities. Twenty-four filled dispensers shall be packed as specified in 5.1.1 except that the box shall conform to style RSC, type CF, class weather-resistant, variety SW, grade W5c of PPP-B-636 and the box shall be closed as specified in the appendix to PPP-B-636.

5.1.2.2 For civil agencies. Twenty-four filled dispensers shall be packed in six rows of four dispensers per row in a close-fitting box conforming to class domestic of PPP-B-636. Slotted partitions shall be used to separate each dispenser. The partitions shall be made of the same material as used for the fiberboard box. The box shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C.

5.1.3.1 For military activities. Packing shall be in accordance with Fed. Std. No. 356.

5.1.3.2 For civil agencies. The aerosol insecticide shall be packed in a container complying with Uniform Freight Classification or National Motor Freight Classification, as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

5.2 Marking. Marking shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for level A and B containers for military activities. Level C packs shall be marked in accordance with Fed. Std. No. 356. Each can and shipping container shall be marked to show date of packaging and lot number. Shipping containers shall also be marked in accordance with Department of Transportation (DOT) regulations, including the DOT green "Nonflammable Compressed Gas" label.

5.3 Palletization. When specified in the contract or order (see 6.2), shipping containers prepared as specified in 5.1 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the softwood pallet conforming to type IV of NN-P-71.

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6. NOTES

6.1 Intended use. Type I aerosol insecticide is intended for indoor use against mosquitoes, flies, and gnats. Type II aerosol insecticide is intended for use aboard aircraft for purposes of public health, plant protection, and quarantine insect control.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Level of packing required (see 5.1).
(d) If palletization is required (see 5.3).

6.3 Water content. In order to minimize water content, it is recommended that filling be carried out a relative humidity equivalent to no more than 30 percent at 70° F.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Analytical standard resmethrin. Analytical standard resmethrin may be obtained from S. B. Penick and Company, Commercial Development - Pesticides, 215 Watchung Avenue, Orange NJ 07050.

6.6 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E29).

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MILITARY INTERESTS:

Custodians:

Army - EA
Navy - AS
Air Force - 68

Review activities:

Army - MD
DSA - GS

User activity:

Navy - SH

Civil Agency Coordinating Activities:

EPA
GSA - FSS
VA - DMS

Preparing activity:

Army - EA

Orders for this publication are to be placed with (s Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

O-I-1925, Insecticide, Resmethrin (Aerosol)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Commanding Officer
Edgewood Arsenal
Attn: SAREA-TS-S
Aberdeen Proving Ground, MD 21010

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